

From the Binomial Theorem to Coq: An Educational Explanation of Version 16.4 of the Conditional Logarithmic Model of Fermat's Last Theorem

G. L. Dedenko

Candidate of Physical and Mathematical Sciences

Senior Lecturer at the Department of Information Technologies (DIT)

Financial University under the Government of the Russian Federation

June 3, 2026

Abstract

This document presents an educational explanation of version 16.4 of the conditional reconstructive model around Fermat's Last Theorem. The text is written for a strong senior high school student of a mathematical lyceum or a first-year university student. It explains not the standard Wiles–Taylor proof, but a separate conditional algebraic-logarithmic scheme. It introduces the Fermat equation, parametrization via m, p , the odd binomial core $S_n(m, p)$, two natural real normalizations of the same core, the exact ratio

$$l^n = q^n \frac{2^{n-1}}{n},$$

the logarithmic gap

$$\Delta_n := \ln \frac{l}{q},$$

the zero logarithmic gap condition $\Delta_n = 0$, the central chain of version 16.4

$$l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1},$$

and the machine-verified final obstruction in Coq

$$n = 2^{n-1} \implies n \in \{1, 2\}.$$

The real, integer, and modular levels are considered separately so as not to confuse equalities over $\mathbb{R}$ and $\mathbb{Z}$ with modular congruences. The main goal of the text is not to announce an unconditional new proof of Fermat's Last Theorem, but to clearly show the conditional logical structure: under two natural normalizations of the same odd binomial core, the zero logarithmic gap condition is compatible with the structure of binomial coefficients only in the linear and quadratic cases. If it is independently justified that a hypothetical positive integer solution must yield $\Delta_n = 0$, then an exponent $n > 2$ is excluded within this conditional model.

Keywords: Fermat's Last Theorem; conditional mathematical model; binomial theorem; odd binomial core; real factorization; two normalizations of the core; residual factors; logarithmic gap; $\Delta_n = \ln(l/q)$; zero-gap condition; logarithmic compatibility obstruction; $l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1}$; Coq; formal verification; GlobalNormalization.v; modular caveat; integer specialization; educational presentation for a mathematical lyceum.

Contents

Preliminary Honest Remark on the Status of the Model	4
Main Idea of Version 16.4	4
1 Historical Formulation of the Problem	5
2 How a Mathematical Lyceum Student Should Read This Document	6
3 Map of Correspondence Between the Article, the Educational Text, and Coq	6
4 Parametrization via m and p	7
5 Three Levels of Numbers: $\mathbb{R}$, $\mathbb{Z}$, the Modular, and Logarithmic Levels	8
5.1 Real Level	8
5.2 Integer Level	8
5.3 Modular Level	9
5.4 Logarithmic Level	9
6 The Binomial Theorem and the Odd Binomial Core	9
7 The First Real Normalization: $S_n(m, p) = n l^n$	10
8 Why This Is Not Integer Divisibility	11
9 The Second Real Normalization: $S_n(m, p) = 2^{n-1} q^n$	11
10 Comparison of the Two Normalizations	12
11 The Logarithmic Gap Δ_n	12
12 The Zero Gap Condition and the Compatibility of Residual Scales	13
13 The Logarithmic Compatibility Obstruction of Coefficient Symmetry	14
14 The Quadratic Case and the Positive Logarithmic Gap for $n > 2$	14
15 Why $n = 2^{n-1}$ Yields Only $n = 1, 2$	16
16 The Conditional Theorem of the Version 16.4 Model	17
17 The Modular Caveat	18
18 How the Modular Caveat is Reflected in Coq	18
19 Connection with the Main Coq Fragments of Version 16.4	19
20 What Exactly Coq Verifies	20
21 What Coq Does Not Prove	20
22 Why Compilation Time is Important But Does Not Replace Mathematics	21
23 How to Correctly Formulate a Message for a Forum	21

24 Frequently Asked Questions	21
24.1 Does this educational file prove FLT unconditionally?	21
24.2 What is the main mathematical knot of version 16.4?	22
24.3 What is Δ_n ?	22
24.4 Why does $\Delta_n = 0$ mean $l^n = q^n$?	22
24.5 What exactly does Coq verify?	22
24.6 Why is q not a modulus?	22
24.7 Why is real factorization not divisibility?	22
24.8 Why is entropy not a proof?	22
24.9 Why is the quadratic case special?	23
24.10 What happens for $n > 2$?	23
25 Educational Exercises	23
26 Final Scheme of the Entire Model	24
27 Conclusion	24
28 Mini-Glossary	25
29 List of Sources and References	26

Preliminary Honest Remark on the Status of the Model

Fermat's Last Theorem has already been proven by modern mathematics in the works of Andrew Wiles and Richard Taylor-Wiles. This educational text does not replace that proof and does not constitute an unconditional new proof of the theorem.

Here, a different task is addressed: to carefully present a conditional reconstructive model in which the odd binomial core is isolated, two natural real normalizations of this core are compared, and then this comparison is rewritten in logarithmic form via

$$\Delta_n = \ln \frac{l}{q}.$$

Therefore, a key distinction must be established right away:

The unconditional proof asserts FLT without additional hypotheses. **The conditional model of version 16.4** shows: under two real normalizations of the same odd binomial core, the zero logarithmic gap condition $\Delta_n = 0$ is equivalent to the equality of residual scales $l^n = q^n$ and the equality $n = 2^{n-1}$. Therefore, under such a compatibility requirement, an exponent $n > 2$ is impossible.

This model consists of two parts:

1. **The algebraic-reconstructive part:** from the symmetric notation of the difference of powers, the odd binomial core $S_n(m, p)$ and its two real normalizations are extracted.
2. **The formally verifiable part:** after accepting the two factorizations, the positivity of l, q , and the condition $\Delta_n = 0$, Coq verifies the chain

$$l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1},$$

and subsequently the final conclusion $n \in \{1, 2\}$.

This separation is important. The computer verification does not replace the mathematical justification that a hypothetical positive integer solution is obliged to generate the condition $\Delta_n = 0$. It merely guarantees that the subsequent logical transitions within the chosen conditional scheme are strictly executed.

The language of entropy, if used in the article, is solely a heuristic interpretation of the logarithmic gap. No physical principle of entropy is used as an independent mathematical axiom or proof.

Main Idea of Version 16.4

The main idea of version 16.4 is that the comparison of two real normalizations of the same odd binomial core can be written in logarithmic form. First, we have

$$\text{core} = n l^n, \quad \text{core} = 2^{n-1} q^n.$$

This implies the exact ratio

$$l^n = q^n \frac{2^{n-1}}{n}.$$

Since $l > 0$ and $q > 0$, we can introduce the logarithmic gap

$$\Delta_n = \ln \frac{l}{q}.$$

Then

$$\Delta_n = \frac{1}{n} \ln \frac{2^{n-1}}{n}.$$

The zero gap condition

$$\Delta_n = 0$$

is equivalent to the equality of residual scales

$$l^n = q^n,$$

and, given the two normalizations of the same core, is equivalent to

$$n = 2^{n-1}.$$

Next, Coq verifies the elementary step

$$n = 2^{n-1} \implies n \in \{1, 2\}.$$

Therefore, under the zero-gap compatibility requirement, an exponent higher than a square is excluded.

For a mathematical lyceum student, this can be formulated as follows: two normalizations of one core create two residual quantities l and q . The logarithm of the ratio l/q measures their divergence. In the quadratic case, this divergence is equal to zero. For every $n > 2$, the coefficient mass 2^{n-1} is greater than the linear coefficient n , and the logarithmic gap becomes positive.

1 Historical Formulation of the Problem

Fermat's Last Theorem states that the equation

$$x^n + y^n = z^n \tag{1}$$

has no solutions in positive integers x, y, z for integer $n > 2$.

For $n = 1$, solutions are obvious: for example, $1 + 2 = 3$. For $n = 2$, Pythagorean triples exist:

$$3^2 + 4^2 = 5^2.$$

But starting with cubes, the situation changes: one cannot represent a cube as the sum of two cubes, a fourth power as the sum of two fourth powers, and so on.

Fermat formulated this statement in the margins of his copy of Diophantus's *Arithmetica* and wrote that he had found a "truly marvelous proof", but the margins of the book were too narrow to contain it. From a historical perspective, it is natural to ask: could Fermat have possessed an elementary idea that he did not write down in modern form?

This document does not claim that Fermat's historical proof has exactly been found. It proposes an educational reconstruction of a conditional scheme showing how a strict implication might arise from the binomial theorem, the symmetry of two terms, the comparison of two real residual factors, and the logarithmic zero-gap condition.

2 How a Mathematical Lyceum Student Should Read This Document

To understand the text, it is sufficient to confidently master the following topics:

- powers, roots, and natural numbers;
- the binomial theorem;
- elementary divisibility;
- logarithms of positive numbers;
- comparison of function growth rates;
- a basic understanding of what formal proof verification is.

The main methodological principle: we will distinguish between four things.

1. **Algebraic identity:** an equality obtained by expanding brackets or changing variables.
2. **Definition:** the introduction of a new object, for example, l , q , or Δ_n .
3. **Model condition:** an additional statement that is required to be accepted or proved separately, for example, $\Delta_n = 0$.
4. **Formal derivation:** a corollary that Coq verifies after all premises are explicitly written out.

This distinction guards against a typical error in amateur proofs: a beautiful observation cannot be imperceptibly turned into a proven theorem. In version 16.4, the central mathematical knot is named directly: it is the status of the zero logarithmic gap requirement $\Delta_n = 0$ and its exact connection with the equality of residual scales $l^n = q^n$ and the equality of coefficient masses $n = 2^{n-1}$.

3 Map of Correspondence Between the Article, the Educational Text, and Coq

So as not to get lost in the notation, it is useful to keep the following map in front of your eyes. The name of the main Coq file is kept as `GlobalNormalization.v` for backward compatibility of the GitHub repository; in version 16.4, its mathematical content corresponds to the logarithmic zero-gap obstruction.

Level	Mathematical meaning	Name in Coq
Real parametrization	If $z = m^n + p^n$, $x = m^n - p^n$, then $z + x = 2m^n$, $z - x = 2p^n$ over $\mathbb{R}$.	<code>sum_diff_from_parameters_R</code>
Integer specialization	The same formula over $\mathbb{Z}$; for integer m, p , parity conditions arise.	<code>sum_diff_from_parameters_Z;</code> <code>parity_condition_Z</code>
Checking the impossibility of integer parameters	If $z + x$ or $z - x$ has the wrong parity, there are no integer m, p in this form.	<code>no_parameters_if_parity_violation;</code> <code>no_parameters_if_odd;</code> <code>no_parameters_for_example</code>
First core normalization	The odd core is written as $\text{core} = n l^n$ over $\mathbb{R}$.	<code>real_core_factorization</code>
Second core normalization	The same core is written as $\text{core} = 2^{n-1} q^n$ over $\mathbb{R}$.	<code>coefficient_mass_factorization</code>
Comparison of two normalizations	From the two normalizations of one core, $l^n = q^n 2^{n-1}/n$ follows.	<code>two_scale_factorizations_relation</code>

Continued on the next page

Level	Mathematical meaning	Name in Coq
Equality of residual scales	The equality $l^n = q^n$.	<code>residual_scale_equality</code>
Logarithmic gap	The standard real logarithm of the ratio l/q : $\Delta_n = \ln(l/q)$.	<code>logarithmic_gap</code>
Zero logarithmic gap	The condition $\ln(l/q) = 0$.	<code>zero_logarithmic_gap</code>
Coefficient side of the gap	The formula $(1/n) \ln(2^{n-1}/n)$.	<code>coefficient_logarithmic_gap</code>
Bridge between scales and the gap	For $l, q > 0$, we have $l^n = q^n \iff \Delta_n = 0$.	<code>residual_scale_equality_</code> <code>implies_zero_logarithmic_gap;</code> <code>zero_logarithmic_gap_implies_</code> <code>residual_scale_equality;</code> <code>residual_scale_equality_iff_</code> <code>zero_logarithmic_gap</code>
Equality of coefficient masses	The equality $n = 2^{n-1}$.	<code>coefficient_mass_equality</code>
Algebraic compatibility	Under two factorizations and $q > 0$, we have $l^n = q^n \iff n = 2^{n-1}$.	<code>coefficient_symmetry_</code> <code>compatibility_iff</code>
Logarithmic compatibility of version 16.4	Under two factorizations and $l, q > 0$, we have $\Delta_n = 0 \iff n = 2^{n-1}$.	<code>logarithmic_coefficient_</code> <code>symmetry_compatibility_iff</code>
Central chain of version 16.4	The full chain $l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1}$.	<code>logarithmic_zero_gap_chain</code>
Packing algebraic data	One structure stores n , the core, l , q , positivity, both factorizations, and the equality of residual scales.	<code>CoefficientSymmetryData</code>
Packing logarithmic data	One structure stores n , the core, l , q , positivity, both factorizations, and the condition $\Delta_n = 0$.	<code>Logarithmic</code>
Coefficient SymmetryData		
Derivation from logarithmic data	From the zero-gap compatibility data follows $n = 2^{n-1}$, then $n = 1$ or $n = 2$.	<code>logarithmic_data_forces_shift;</code> <code>logarithmic_data_forces_</code> <code>small_exponent</code>
Exclusion of large exponents	The logarithmic compatibility data and $n > 2$ imply a contradiction.	<code>logarithmic_data_excludes_</code> <code>high_exponent;</code> <code>logarithmic_zero_gap_</code> <code>excludes_high_exponent</code>
Final growth	From $n = 2^{n-1}$ follows $n = 1$ or $n = 2$.	<code>binary_scaling_roots_only_</code> <code>one_two</code>
Modular caveat	Integer equality implies modular congruence, but the converse is not true.	<code>ModularRemark;</code> <code>modular_congruence_not_</code> <code>integer_equality</code>

The educational text explains the idea in human language. The article fixes the mathematical formulation. The Coq file verifies only those implications that are explicitly written in the code.

4 Parametrization via m and p

Suppose hypothetically there exists a positive natural solution to the Fermat equation for some $n > 2$:

$$x^n + y^n = z^n.$$

We rewrite it as

$$z^n - x^n = y^n. \tag{2}$$

Now we introduce parameters m and p using the system

$$\begin{cases} m^n + p^n = z, \\ m^n - p^n = x. \end{cases} \quad (3)$$

Adding and subtracting these equations:

$$2m^n = z + x, \quad 2p^n = z - x.$$

From this, formally:

$$m = \sqrt[n]{\frac{z+x}{2}}, \quad p = \sqrt[n]{\frac{z-x}{2}}. \quad (4)$$

Remark 4.1. If m and p are allowed as real numbers, such a reconstruction is an algebraic notation after choosing a suitable branch of the root. However, if we additionally require $m, p \in \mathbb{Z}$, conditions arise: $z+x$ and $z-x$ must be even, and the numbers $(z+x)/2$ and $(z-x)/2$ must be exact n -th powers in $\mathbb{Z}$.

Substituting (3) into (2), we obtain

$$y^n = (m^n + p^n)^n - (m^n - p^n)^n. \quad (5)$$

This is exactly where the binomial part begins. This step does not claim that the real parametrization automatically yields an integer solution: integer compatibility is considered separately.

5 Three Levels of Numbers: $\mathbb{R}$, $\mathbb{Z}$, the Modular, and Logarithmic Levels

The same expression can exist in different mathematical worlds. In this model, it is important to distinguish four levels.

5.1 Real Level

Here $m, p, l, q \in \mathbb{R}$. We can extract roots, introduce real residual factors, and write down factorizations of the form

$$S_n(m, p) = n l^n, \quad S_n(m, p) = 2^{n-1} q^n.$$

At this level, the notation means equality of real quantities.

5.2 Integer Level

Here we additionally discuss the conditions $x, y, z \in \mathbb{N}$ and, if required, $m, p \in \mathbb{Z}$. For example, from

$$z = m^n + p^n, \quad x = m^n - p^n$$

for $m, p \in \mathbb{Z}$ it follows that

$$z + x = 2m^n, \quad z - x = 2p^n.$$

Consequently, $z+x$ and $z-x$ must be even. If this is not the case, there can be no integer parameters m, p in this form.

Example 5.1. Let $n = 3$, $z = 2$, $x = 1$. Then

$$\frac{z+x}{2} = \frac{3}{2}, \quad \frac{z-x}{2} = \frac{1}{2}.$$

The real roots exist, but there are no integer m, p satisfying

$$2 = m^3 + p^3, \quad 1 = m^3 - p^3.$$

It is exactly this kind of distinction that the lemma `no_parameters_for_example` formalizes.

5.3 Modular Level

The modular level considers remainders. To avoid confusing the modulus with the real residual factor q , we will denote the modulus by `modq`:

$$x^n + y^n \equiv z^n \pmod{\text{modq}}.$$

Such a congruence can be true even when the equality $x^n + y^n = z^n$ in integers is false. Therefore, modular examples by themselves are not counterexamples to FLT.

5.4 Logarithmic Level

After the real level, version 16.4 introduces an additional logarithmic layer. Since l and q are positive, the ratio l/q can be taken inside a logarithm:

$$\Delta_n = \ln \frac{l}{q}.$$

This layer is not modular arithmetic and is not physical entropy. It is the standard real logarithmic notation of the ratio of two positive quantities.

6 The Binomial Theorem and the Odd Binomial Core

Recall the formula for the Binomial Theorem:

$$(A + B)^n = \sum_{r=0}^n \binom{n}{r} A^{n-r} B^r. \quad (6)$$

For the difference we have

$$(A - B)^n = \sum_{r=0}^n (-1)^r \binom{n}{r} A^{n-r} B^r. \quad (7)$$

Subtracting (7) from (6), we get

$$(A + B)^n - (A - B)^n = \sum_{r=0}^n (1 - (-1)^r) \binom{n}{r} A^{n-r} B^r \quad (8)$$

$$= 2 \sum_{\substack{1 \leq r \leq n \\ r \text{ odd}}} \binom{n}{r} A^{n-r} B^r. \quad (9)$$

In our case

$$A = m^n, \quad B = p^n.$$

Therefore

$$y^n = 2 \sum_{\substack{1 \leq r \leq n \\ r \text{ odd}}} \binom{n}{r} (m^n)^{n-r} (p^n)^r. \quad (10)$$

Definition 6.1 (Odd Binomial Core). We will call the sum

$$S_n(m, p) = \sum_{\substack{1 \leq r \leq n \\ r \text{ odd}}} \binom{n}{r} (m^n)^{n-r} (p^n)^r \quad (11)$$

the odd binomial core. Then

$$(m^n + p^n)^n - (m^n - p^n)^n = 2S_n(m, p), \quad (12)$$

and, consequently,

$$y^n = 2S_n(m, p).$$

Intuitively, $S_n(m, p)$ is the part of the binomial expansion that survives after subtracting two symmetric powers. Even terms are canceled out, odd terms are doubled.

7 The First Real Normalization: $S_n(m, p) = n l^n$

For each odd index r , the binomial coefficient can be written as

$$\binom{n}{r} = \frac{n}{r} \binom{n-1}{r-1}. \quad (13)$$

This allows us, in real-rational algebra, to factor out a common multiplier n from the odd core.

Therefore, the first normalization is introduced:

$$S_n(m, p) = n l^n. \quad (14)$$

Here $l > 0$ is a real residual factor. It shows what part of the core remains after factoring out the linear multiplier n .

If $n > 0$, then formally

$$l^n = \frac{S_n(m, p)}{n}. \quad (15)$$

Given a positive expression on the right, we can take the positive real branch of the root.

From $y^n = 2S_n(m, p)$ and (14) we get

$$y^n = 2n l^n. \quad (16)$$

In Coq this corresponds to the definition `real_core_factorization`. It means:

$$0 < n, \quad \text{core} = n l^n$$

over the real numbers.

8 Why This Is Not Integer Divisibility

The notation

$$S_n(m, p) = n l^n$$

over $\mathbb{R}$ does not automatically mean

$$n \mid S_n(m, p)$$

in $\mathbb{Z}$.

This is a fundamentally important distinction. The first statement is a real definition of the residual factor l . The second statement is about integer divisibility, which requires a separate proof and in general might be false.

Example 8.1 (Check for $n = 6$). For $n = 6$, $m = 1$, $p = 1$, the odd sum of binomial coefficients equals

$$\binom{6}{1} + \binom{6}{3} + \binom{6}{5} = 6 + 20 + 6 = 32.$$

The number 32 is not divisible by 6 in $\mathbb{Z}$. But over $\mathbb{R}$ we can write

$$32 = 6 \cdot \frac{16}{3} = 6l^6,$$

if we set

$$l^6 = \frac{16}{3}.$$

This is not a problem for the real model because it uses equality over $\mathbb{R}$, not a statement about divisibility in the ring of integers.

Therefore, the correct formulation is:

$$l^n = \frac{S_n(m, p)}{n} \quad \text{as a real definition,}$$

but not necessarily

$$n \mid S_n(m, p) \quad \text{as an integer statement.}$$

This very distinction is particularly important in version 16.4, since the logarithmic gap

$$\Delta_n = \ln \frac{l}{q}$$

is introduced over positive real residual factors $l > 0$, $q > 0$, and not over integer divisors.

9 The Second Real Normalization: $S_n(m, p) = 2^{n-1}q^n$

Now let's look not at the multiplier n , but at the total mass of the odd binomial coefficients.

From the binomial identity it is known that:

$$\sum_{\substack{1 \leq r \leq n \\ r \text{ odd}}} \binom{n}{r} = 2^{n-1}. \quad (17)$$

This means: the sum of odd coefficients in a row of Pascal's triangle is equal to half the total sum of coefficients.

Therefore, we can introduce a second positive real residual factor $q > 0$:

$$S_n(m, p) = 2^{n-1}q^n. \quad (18)$$

Equivalently,

$$q^n = \frac{S_n(m, p)}{2^{n-1}}. \quad (19)$$

Here q is neither a modulus nor a remainder modulo something. It is a real residual factor that remains after factoring out the mass of odd binomial coefficients 2^{n-1} .

From $y^n = 2S_n(m, p)$ and (18) we obtain

$$y^n = 2^n q^n. \quad (20)$$

In Coq this normalization corresponds to the definition `coefficient_mass_factorization`. Its meaning is:

$$0 < n, \quad \text{core} = 2^{n-1}q^n$$

over the real numbers.

10 Comparison of the Two Normalizations

Now the same core $S_n(m, p)$ is written in two ways:

$$S_n(m, p) = n l^n, \quad S_n(m, p) = 2^{n-1}q^n. \quad (21)$$

Equating the right sides, we get

$$n l^n = 2^{n-1}q^n.$$

Since $n > 0$, we can divide by n :

$$l^n = q^n \frac{2^{n-1}}{n}. \quad (22)$$

This is not a hypothesis, but a direct consequence of two real factorizations of the same core. This is exactly what the Coq lemma `two_scale_factorizations_relation` verifies.

Substantively, formula (22) says: two normalizations of the same core generally yield different residual factors. Their ratio is governed by the coefficient multiplier

$$\frac{2^{n-1}}{n}.$$

11 The Logarithmic Gap Δ_n

Since $l > 0$ and $q > 0$, we can take logarithms in relation (22). From

$$l^n = q^n \frac{2^{n-1}}{n}$$

we get

$$n \ln l = n \ln q + (n-1) \ln 2 - \ln n.$$

From here

$$\ln l - \ln q = \frac{1}{n} \ln \frac{2^{n-1}}{n},$$

that is

$$\ln \frac{l}{q} = \frac{1}{n} \ln \frac{2^{n-1}}{n}. \quad (23)$$

Definition 11.1 (Logarithmic gap). We define the logarithmic gap between the two residual factors as the quantity

$$\Delta_n := \ln \frac{l}{q}. \quad (24)$$

Then from (23) it follows that

$$\Delta_n = \frac{1}{n} \ln \frac{2^{n-1}}{n}. \quad (25)$$

Educationally, this can be understood as follows: Δ_n shows how far the two residual factors l and q have diverged after two different normalizations of the same core. If $\Delta_n = 0$, then $l/q = 1$, meaning $l = q$. If $\Delta_n > 0$, then $l > q$. If $\Delta_n < 0$, then $l < q$.

Sometimes such a quantity can figuratively be called a "gap of an entropy type" because the logarithm of a ratio resembles the language of informational comparisons. But in a strict educational text, it is merely the logarithm of the ratio of two positive real factors. No physical entropy proves anything here.

12 The Zero Gap Condition and the Compatibility of Residual Scales

The zero logarithmic gap condition takes the form

$$\Delta_n = 0.$$

By definition, this means

$$\ln \frac{l}{q} = 0.$$

Since $l/q > 0$, the equality $\ln(l/q) = 0$ is equivalent to

$$\frac{l}{q} = 1,$$

that is

$$l = q.$$

And since $l, q > 0$, the equality $l = q$ is equivalent to the equality of their n -th powers:

$$l^n = q^n.$$

Therefore, for $l > 0$, $q > 0$, $n > 0$ we have

$$\Delta_n = 0 \iff l^n = q^n. \quad (26)$$

In Coq, this is reflected by three statements:

- `residual_scale_equality_implies_zero_logarithmic_gap`;
- `zero_logarithmic_gap_implies_residual_scale_equality`;
- `residual_scale_equality_iff_zero_logarithmic_gap`.

13 The Logarithmic Compatibility Obstruction of Coefficient Symmetry

After comparing the two normalizations, we obtained the exact ratio

$$l^n = q^n \frac{2^{n-1}}{n}.$$

In logarithmic form, it is written as

$$\Delta_n = \frac{1}{n} \ln \frac{2^{n-1}}{n}.$$

Therefore, the condition $\Delta_n = 0$ is equivalent to

$$\ln \frac{2^{n-1}}{n} = 0,$$

which is equivalent to

$$\frac{2^{n-1}}{n} = 1,$$

that is

$$n = 2^{n-1}.$$

On the other hand, from the previous section

$$\Delta_n = 0 \iff l^n = q^n.$$

Consequently, under two real normalizations of the same odd binomial core, we get the central chain of version 16.4:

$$l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1}. \quad (27)$$

For a lyceum student, this can be read as follows: the two residual quantities coincide exactly when the logarithmic gap between them is equal to zero; and this, under two normalizations of the same core, is possible exactly when the two extracted coefficient masses coincide: the linear coefficient n and the total mass of the odd binomial coefficients 2^{n-1} .

In Coq, the algebraic core of this thought is given by the theorem `coefficient_symmetry_compatibility_iff`. The logarithmic enhancement of version 16.4 is given by the theorems `logarithmic_coefficient_symmetry_compatibility_iff` and `logarithmic_zero_gap_chain`.

14 The Quadratic Case and the Positive Logarithmic Gap for $n > 2$

Let's look at what the formula

$$\Delta_n = \frac{1}{n} \ln \frac{2^{n-1}}{n}$$

means for the first few values of n .

The Quadratic Case

For $n = 2$, the corresponding row of Pascal's triangle is

$$1, 2, 1.$$

In the difference of two symmetric squares, only one odd binomial coefficient remains:

$$\binom{2}{1} = 2.$$

The first normalization extracts

$$n = 2,$$

and the second normalization extracts

$$2^{2-1} = 2.$$

Both coefficient masses coincide, therefore

$$\Delta_2 = \frac{1}{2} \ln \frac{2}{2} = 0.$$

This means the two residual factors can coincide:

$$l^2 = q^2.$$

This is exactly the quadratic, Pythagorean case.

The Cubic Case

For $n = 3$, the odd binomial coefficients are

$$\binom{3}{1} = 3, \quad \binom{3}{3} = 1.$$

Their total mass is

$$2^{3-1} = 4.$$

The first normalization extracts 3, and the second extracts 4. Therefore

$$l^3 = q^3 \frac{4}{3},$$

and the logarithmic gap is

$$\Delta_3 = \frac{1}{3} \ln \frac{4}{3} > 0.$$

The Fourth Power

For $n = 4$, the odd coefficients are

$$\binom{4}{1} = 4, \quad \binom{4}{3} = 4.$$

Their total mass is

$$2^{4-1} = 8.$$

The first normalization extracts 4, and the second extracts 8. Therefore

$$l^4 = 2q^4,$$

and

$$\Delta_4 = \frac{1}{4} \ln 2 > 0.$$

In general, for every integer $n > 2$, it holds that

$$2^{n-1} > n.$$

Consequently,

$$\Delta_n = \frac{1}{n} \ln \frac{2^{n-1}}{n} > 0.$$

This is precisely what is called the logarithmic compatibility obstruction: the zero gap condition $\Delta_n = 0$ is compatible with the two normalizations only for $n = 1$ and $n = 2$.

15 Why $n = 2^{n-1}$ Yields Only $n = 1, 2$

Consider the equation

$$n = 2^{n-1}.$$

For $n = 1$, we get

$$1 = 2^0 = 1.$$

For $n = 2$, we get

$$2 = 2^1 = 2.$$

That is, $n = 1$ and $n = 2$ are indeed solutions.

Let us show that there are no solutions for $n > 2$. Equivalently, we can write

$$2^n = 2n.$$

But for all natural numbers $n \geq 3$, we have

$$2^n > 2n.$$

For example:

$$2^3 = 8 > 6, \quad 2^4 = 16 > 8, \quad 2^5 = 32 > 10.$$

Further on, the gap only grows, because the exponential function doubles at each step, while the linear function only adds a constant value.

In Coq language, this part is split into several elementary lemmas:

- `pow2_gt_linear_shift`;
- `pow2_gt_linear`;
- `pow_eq_linear_positive`;
- `two_pow_eq_two_mul_iff_shift`;
- `binary_scaling_roots_only_one_two`.

The overall formal thought is simple:

$$n = 2^{n-1} \implies n = 1 \text{ or } n = 2.$$

In version 16.4, this is used as the end of the logarithmic chain:

$$\Delta_n = 0 \implies n = 2^{n-1} \implies n \in \{1, 2\}.$$

16 The Conditional Theorem of the Version 16.4 Model

Theorem 16.1 (Conditionally under a zero logarithmic gap). *Let a hypothetical positive integer solution to the Fermat equation*

$$x^n + y^n = z^n, \quad n > 2,$$

admit a real symmetric parametrization

$$z = m^n + p^n, \quad x = m^n - p^n.$$

Let the odd binomial core $S_n(m, p)$ admit two real normalizations

$$S_n(m, p) = n l^n, \quad S_n(m, p) = 2^{n-1} q^n,$$

where $l > 0, q > 0$. Then a comparison of the two normalizations yields

$$l^n = q^n \frac{2^{n-1}}{n}.$$

Introducing

$$\Delta_n = \ln \frac{l}{q},$$

we obtain

$$l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1}.$$

Since the equation $n = 2^{n-1}$ has only the solutions $n = 1$ and $n = 2$ among positive integers, the zero logarithmic gap condition is incompatible with any exponent $n > 2$.

Proof. From the two normalizations it follows that

$$n l^n = 2^{n-1} q^n,$$

which means

$$l^n = q^n \frac{2^{n-1}}{n}.$$

Since $l > 0$ and $q > 0$, we can take logarithms and get

$$\Delta_n = \frac{1}{n} \ln \frac{2^{n-1}}{n}.$$

If $\Delta_n = 0$, then $\ln(l/q) = 0$, hence $l = q$, which means $l^n = q^n$. Conversely, if $l^n = q^n$ for $l, q > 0$, then $l = q$, hence $\Delta_n = 0$. Given the two normalizations, this is equivalent to $n = 2^{n-1}$. But this equation has only the solutions $n = 1$ and $n = 2$ in positive integers. Therefore, the case $n > 2$ contradicts the requirement of a zero logarithmic gap. $\square$

This theorem is conditional. It does not prove that every hypothetical positive integer solution is obliged to yield $\Delta_n = 0$. That exactly is the main mathematical question of the model.

17 The Modular Caveat

In discussions of FLT, examples of congruences of the form

$$a^n + b^n \equiv c^n \pmod{\text{mod}q}$$

are often cited. Such examples may be correct, but they belong to a different level of the problem.

A congruence modulo something means that there exists an integer t such that

$$c^n = a^n + b^n + \text{mod}q \cdot t.$$

Equality in integers is only a special case where $t = 0$.

Example 17.1. We have

$$1^3 + 1^3 = 2, \quad 3^3 = 27.$$

In integers $2 \neq 27$. But modulo 5

$$27 = 2 + 5 \cdot 5,$$

meaning

$$3^3 \equiv 1^3 + 1^3 \pmod{5}.$$

This is a true modular congruence, but not a true integer equality.

Therefore, the existence of solutions to congruences

$$x^n + y^n \equiv z^n \pmod{\text{mod}q}$$

for $n > 2$ is not a counterexample to Fermat's Last Theorem and does not refute the real parametrization.

If one specifically transitions to the modular level, the parameters must be set separately:

$$m^n \equiv A \pmod{\text{mod}q}, \quad p^n \equiv B \pmod{\text{mod}q},$$

and only then can one write

$$z \equiv A + B \pmod{\text{mod}q}, \quad x \equiv A - B \pmod{\text{mod}q}.$$

In version 16.4, this modular caveat retains its previous meaning. It does not participate directly in the logarithmic zero-gap obstruction, but it protects the reasoning from confusing integer equalities with modular congruences.

18 How the Modular Caveat is Reflected in Coq

In Coq, there is a section `ModularRemark`. In it, the relation of modular congruence is defined as

$$\text{modular_congruent mod}q \ a \ b.$$

Mathematically, this means there exists an integer t such that

$$a = b + \text{mod}q \cdot t.$$

The lemma `integer_equality_implies_modular_power_congruence` verifies the direction

$$\text{integer equality} \implies \text{modular congruence}.$$

If $x^n + y^n = z^n$, then one can take $t = 0$ and obtain the corresponding congruence for any modulus $\text{mod } q$.

The lemma `integer_equality_is_zero_modular_witness` explicitly fixes that a true equality in integers corresponds to a zero modular witness $t = 0$.

The lemma `modular_congruence_not_integer_equality` verifies a specific example:

$$3^3 \equiv 1^3 + 1^3 \pmod{5},$$

since $27 = 2 + 5 \cdot 5$, but at the same time

$$1^3 + 1^3 \neq 3^3$$

as an equality in integers.

The structure `ModularResidues` formalizes the idea that on the modular level, one must separately define the residues m^n and p^n , and then separately define the residues z and x .

19 Connection with the Main Coq Fragments of Version 16.4

The old algebraic core of version 15.3 is retained in Coq. This includes:

- `real_core_factorization`;
- `coefficient_mass_factorization`;
- `two_scale_factorizations_relation`;
- `residual_scale_equality`;
- `coefficient_mass_equality`;
- `coefficient_symmetry_compatibility_iff`;
- `CoefficientSymmetryData`.

But version 16.4 adds a logarithmic layer:

- `logarithmic_gap` means $\ln(l/q)$;
- `zero_logarithmic_gap` means $\ln(l/q) = 0$;
- `coefficient_logarithmic_gap` sets the coefficient side of the gap $(1/n) \ln(2^{n-1}/n)$;
- `residual_scale_equality_implies_zero_logarithmic_gap` derives the zero gap from the equality $l^n = q^n$;
- `zero_logarithmic_gap_implies_residual_scale_equality` derives $l^n = q^n$ from the zero gap;
- `residual_scale_equality_iff_zero_logarithmic_gap` combines the two previous directions;
- `logarithmic_coefficient_symmetry_compatibility_iff` connects $\Delta_n = 0$ with $n = 2^{n-1}$;
- `logarithmic_zero_gap_chain` formalizes the chain $l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1}$;
- `LogarithmicCoefficientSymmetryData` packs the two factorizations, positivity of l, q , and the zero-gap condition;
- `logarithmic_data_forces_shift` derives $n = 2^{n-1}$;

- `logarithmic_data_forces_small_exponent` derives $n = 1$ or $n = 2$;
- `logarithmic_data_excludes_high_exponent` excludes $n > 2$ for packed zero-gap data;
- `logarithmic_zero_gap_excludes_high_exponent` formulates the final exclusion of $n > 2$ directly from the two factorizations, positivity of l, q , and the condition $\Delta_n = 0$.

The main thing version 16.4 does: it does not throw away the old equality of residual scales, but places it into a more transparent chain via the logarithmic gap.

20 What Exactly Coq Verifies

The Coq file of version 16.4 verifies the following conditional algebraic-logarithmic chain.

1. Real parametric identities:

$$z = m^n + p^n, \quad x = m^n - p^n \implies z + x = 2m^n, \quad z - x = 2p^n.$$

2. Integer parity conditions for $m, p \in \mathbb{Z}$.
3. The distinction between a modular congruence and an integer equality.
4. The meaning of the two real factorizations:

$$\text{core} = n l^n, \quad \text{core} = 2^{n-1} q^n.$$

5. The exact ratio:

$$l^n = q^n \frac{2^{n-1}}{n}.$$

6. The logarithmic bridge:

$$l^n = q^n \iff \Delta_n = 0.$$

7. Compatibility with the coefficient mass:

$$\Delta_n = 0 \iff n = 2^{n-1}.$$

8. The final obstruction:

$$n = 2^{n-1} \implies n \in \{1, 2\}.$$

9. Exclusion of $n > 2$ under the condition `zero_logarithmic_gap`.

That is, Coq verifies not the emotional or historical part, but only the formalized conditional chain.

21 What Coq Does Not Prove

Coq does not prove that every hypothetical positive integer solution to the Fermat equation is obliged to generate the condition

$$\Delta_n = 0.$$

Coq also does not prove the historical fact that Fermat reasoned exactly in this way. And Coq does not prove any physical or entropy principle.

Coq verifies only the conditional mathematical chain:

$$\begin{aligned} & \text{two real factorizations} + l > 0 + q > 0 + \Delta_n = 0 \\ \implies & n = 2^{n-1} \implies n \in \{1, 2\}. \end{aligned}$$

Therefore, the main subject of criticism for version 16.4 is the mathematical status of the zero-gap compatibility requirement. In other words, one must discuss not whether Coq correctly derives $n \in \{1, 2\}$ from $n = 2^{n-1}$, but why a hypothetical positive integer solution would have to yield $\Delta_n = 0$.

22 Why Compilation Time is Important But Does Not Replace Mathematics

The compilation time shows that the Coq file is technically verified. In the current file of version 16.4 it is stated:

Calculation time: 3.1s

This is a good technical sign: formal definitions and lemmas are consistent, the syntax is correct, and the proofs pass through the verifying kernel of Coq.

But compilation time does not mean that Coq has proved FLT unconditionally. It only means that the formalized lemmas and theorems inside the conditional model are consistent and pass machine verification.

23 How to Correctly Formulate a Message for a Forum

If discussing this version on a mathematical forum, a correct formulation must be cautious. For example:

I do not claim that Coq has proven FLT unconditionally. I propose a conditional reconstruction. In it, the same odd binomial core has two real normalizations. Their comparison yields a logarithmic gap $\Delta_n = \ln(l/q)$. Under the condition $\Delta_n = 0$, we get $n = 2^{n-1}$, and Coq verifies that this implies $n = 1$ or $n = 2$. The main question is whether the zero-gap compatibility requirement is mathematically justified for a hypothetical integer solution.

Such a formulation honestly separates what is proved inside the model from what requires additional justification.

24 Frequently Asked Questions

24.1 Does this educational file prove FLT unconditionally?

No. It explains the conditional reconstructive model of version 16.4. The unconditional proof of FLT remains the Wiles-Taylor proof.

24.2 What is the main mathematical knot of version 16.4?

The main knot is the status of the zero-gap compatibility requirement: is a hypothetical positive integer solution obliged to yield

$$\Delta_n = 0?$$

If yes, then the exponent $n > 2$ is excluded. If not, the model remains only a conditional reconstruction.

24.3 What is Δ_n ?

It is the logarithmic gap between two positive real residual factors:

$$\Delta_n = \ln \frac{l}{q}.$$

It shows whether the two residual factors coincide after two different normalizations of the same core.

24.4 Why does $\Delta_n = 0$ mean $l^n = q^n$?

If $\Delta_n = 0$, then $\ln(l/q) = 0$. Since $l/q > 0$, we get $l/q = 1$, that is $l = q$, which means $l^n = q^n$. Conversely, if $l^n = q^n$ for $l, q > 0$, then $l = q$, and therefore $\Delta_n = 0$.

24.5 What exactly does Coq verify?

Coq verifies the conditional chain of the two factorizations, the zero-gap condition, and the final exclusion of $n > 2$:

$$\text{core} = n l^n, \quad \text{core} = 2^{n-1} q^n, \quad \Delta_n = 0 \quad \implies \quad n \in \{1, 2\}.$$

24.6 Why is q not a modulus?

Because q in this model is a positive real residual factor. A modulus in modular arithmetic is denoted separately as $\text{mod } q$.

24.7 Why is real factorization not divisibility?

Because the notation $\text{core} = n l^n$ over $\mathbb{R}$ means a real equality. It does not automatically mean $n \mid \text{core}$ in $\mathbb{Z}$.

24.8 Why is entropy not a proof?

The language of entropy is only a heuristic analogy. The strict part of version 16.4 is the algebraic-logarithmic chain through $\Delta_n = \ln(l/q)$.

24.9 Why is the quadratic case special?

For $n = 2$, we have

$$2^{n-1} = 2 = n.$$

Therefore

$$\Delta_2 = \frac{1}{2} \ln \frac{2}{2} = 0.$$

This is exactly why the quadratic case is compatible with the equality of residual scales.

24.10 What happens for $n > 2$?

For each $n > 2$, we have

$$2^{n-1} > n.$$

Therefore

$$\Delta_n = \frac{1}{n} \ln \frac{2^{n-1}}{n} > 0.$$

Meaning, the condition of a zero logarithmic gap is impossible.

25 Educational Exercises

1. Verify that for $n = 2$ the value of Δ_2 is zero.
2. Calculate $\Delta_3 = \frac{1}{3} \ln \frac{4}{3}$ and explain why it is positive.
3. Calculate $\Delta_4 = \frac{1}{4} \ln 2$ and explain why it is positive.
4. Prove elementarily that $2^{n-1} > n$ for all $n > 2$.
5. Explain why $\ln(l/q) = 0$ for $l, q > 0$ is equivalent to $l = q$.
6. Explain why $l = q$ for $l, q > 0$ is equivalent to $l^n = q^n$.
7. Using the example $n = 6, m = p = 1$, explain why real factorization is not integer divisibility.
8. Using the example $3^3 \equiv 1^3 + 1^3 \pmod{5}$, explain why a modular congruence is not an integer equality.
9. Find the definitions of `logarithmic_gap` and `zero_logarithmic_gap` in the Coq file and explain them in plain words.
10. Find the theorem `logarithmic_zero_gap_excludes_high_exponent` in the Coq file and retell it without Coq syntax.

26 Final Scheme of the Entire Model

In a condensed form, the entire conditional scheme looks like this:

$$\begin{aligned}
 & x^n + y^n = z^n \\
 & \Downarrow \\
 & y^n = (m^n + p^n)^n - (m^n - p^n)^n \\
 & \Downarrow \\
 & (m^n + p^n)^n - (m^n - p^n)^n = 2S_n(m, p) \\
 & \Downarrow \\
 & S_n(m, p) = n l^n, \quad S_n(m, p) = 2^{n-1} q^n \\
 & \Downarrow \\
 & l^n = q^n \frac{2^{n-1}}{n} \\
 & \Downarrow \\
 & \Delta_n := \ln \frac{l}{q} = \frac{1}{n} \ln \frac{2^{n-1}}{n} \\
 & \Downarrow \\
 & l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1} \\
 & \Downarrow \\
 & n \in \{1, 2\} \\
 & \Downarrow \\
 & \text{for } n > 2 \text{ the condition of a zero logarithmic gap} \\
 & \quad \text{is incompatible with the two normalizations.}
 \end{aligned}$$

27 Conclusion

Version 16.4 makes the educational scheme more transparent: the previous equality of residual scales no longer hangs on its own, but is incorporated into a logarithmic chain via Δ_n . The strict formal part takes the form

$$l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1} \implies n \in \{1, 2\}.$$

Thereby, the binomial symmetry of coefficients shows: in the quadratic case, the logarithmic gap vanishes, and for every exponent $n > 2$, it becomes strictly positive. This neatly explains why the linear and quadratic cases are consistent with a zero gap, whereas higher exponents are not.

The main conditional point remains the same in meaning, but is formulated more accurately: it is necessary to justify why a hypothetical positive integer solution is obliged to yield $\Delta_n = 0$. Until this is independently proven, version 16.4 is a conditional reconstruction, and not an unconditional proof of Fermat's Last Theorem.

The educational significance of the document is that it highlights the boundary between an idea, a definition, a model condition, a formal derivation, and machine verification. Such a boundary is especially important when discussing strong, yet controversial mathematical reconstructions.

28 Mini-Glossary

FLT	Fermat's Last Theorem: the equation $x^n + y^n = z^n$ has no positive integer solutions for $n > 2$.
$S_n(m, p)$	The odd binomial core, i.e., the sum of the odd terms in the difference $(m^n + p^n)^n - (m^n - p^n)^n$ after division by 2.
l	A positive real residual factor after factoring out the linear coefficient n .
q	A positive real residual factor after factoring out the total mass of the odd binomial coefficients 2^{n-1} ; not a modulus.
Δ_n	The logarithmic gap: $\Delta_n = \ln(l/q)$.
zero-gap condition	The zero logarithmic gap condition: $\Delta_n = 0$.
zero-gap compatibility requirement	The requirement that in the considered conditional model the logarithmic gap must be zero.
Logarithmic compatibility obstruction	The fact that under two normalizations $\Delta_n = 0$ is equivalent to $n = 2^{n-1}$ and therefore impossible for $n > 2$.
logarithmic_gap	The Coq definition of the logarithmic gap.
zero_logarithmic_gap	The Coq predicate signifying a zero logarithmic gap.
LogarithmicCoefficientSymmetryData	A Coq structure packing the data of the two factorizations, the positivity of l, q , and the zero-gap condition.
logarithmic_zero_gap_chain	A Coq theorem formalizing the chain $l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1}$.
GlobalNormalization.v	The main name of the Coq file, retained for backward compatibility of the repository. In version 16.4, its content corresponds to the logarithmic zero-gap obstruction.
Entropy-type gap	A heuristic image of the logarithmic gap; not a mathematical axiom and not a proof.

29 List of Sources and References

1. G. L. Dedenko. *The “Difficulties” in Fermat’s Original Discourse on the Indecomposability of Powers Greater Than a Square: A Retrospect*. Main article, version 16.4.
2. G. L. Dedenko. *“Sharp Corners” in Pierre de Fermat’s Original Reasoning on the Indecomposability of Powers Greater Than a Square: A Review*. Russian version of article 16.4.
3. GlobalNormalization.v. Coq formalization, version 16.4. The file name is retained for backward compatibility; the content corresponds to the logarithmic zero-gap obstruction.
4. *On the Correctness of Real Parametrization, the Modular Caveat, and the Logarithmic Compatibility Obstruction of Coefficient Symmetry*. Explanation for the Coq formalization of version 16.4.
5. Andrew Wiles. *Modular elliptic curves and Fermat’s Last Theorem*. Annals of Mathematics, 1995.
6. Richard Taylor and Andrew Wiles. *Ring-theoretic properties of certain Hecke algebras*. Annals of Mathematics, 1995.
7. Educational materials on the Binomial Theorem, Pascal’s triangle, and the elementary comparison of function growth rates.
8. GitHub Repository FLT-Coq: Coq file GlobalNormalization.v and related materials.